



Silibinin attenuates 3-nitropropionic acid-induced ovarian toxicity by alleviating oxidative stress and granulosa cell apoptosis

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ABSTRACT

3-Nitropropionic acid (3-NP), a mycotoxin present in various plants and fungi, poses significant reproductive health risks to animals and humans through food chains contamination. This study aimed to explore the protective effects and mechanisms of silibinin, a bioactive flavonoid derived from the herbal plant *Silybum marianum*, against 3-NP-induced reproductive toxicity. Our findings demonstrated that silibinin treatment significantly alleviated 3-NP-induced ovarian follicular atresia and preserved ovarian histoarchitecture. Furthermore, it attenuated oxidative stress induced by 3-NP. Molecular analyses through qPCR and Western blot revealed that silibinin upregulated *Bax* and *Caspase-3* expressions while downregulating *Bcl-2* expression in the ovaries compared to the 3-NP group. Immunohistochemistry analysis of CASPASE-3 demonstrated that silibinin significantly inhibited 3-NP-induced apoptosis, predominantly in granulosa cells. Additionally, Western blot analyses showed that silibinin suppressed 3-NP-induced activation of JNK and ERK phosphorylation, down-regulated KEAP1 expression, and upregulated NRF2 nuclear translocation in the ovary. Overall, our results indicate that silibinin effectively alleviated 3-NP-induced ovarian oxidative damage by modulating the JNK/ERK signaling pathways and activating the KEAP1/NRF2 signaling pathway. These findings suggest that silibinin may have potential therapeutic application for mitigating 3-NP-induced reproductive toxicity in animal husbandry and the veterinary industry.

1. Introduction

3-Nitropropionic acid (3-NP), a natural occurring mycotoxin produced by diverse fungi and herbaceous plants, poses significant health risks to animals and humans health through food chain contamination [1,2]. 3-Succinate dehydrogenase (SDH), a key mitochondrial membrane enzyme, serves a crucial role in connecting the respiratory chain to the citric acid cycle [3,4]. Structurally similar to succinate, 3-NP acts as an irreversible inhibitor of mitochondrial complex II by specifically binding to SDH [4–6]. This inhibition SDH disrupts mitochondrial function, leading to reduced ATP production, reactive oxygen species (ROS) accumulation, oxidative stress, and apoptosis [7]. As a well-known neurotoxic compound, 3-NP is widely used neurological dysfunction in experimental animals [3,8]. Furthermore, increasing evidences indicate that 3-NP exhibits significant reproductive toxicity and diminishes reproductive capacity [9,10]. Specifically, 3-NP induces

ROS production in mouse ovarian tissues, leading to granulosa cell apoptosis and follicular atresia [11,12]. Recent research further demonstrates that 3-NP causes mitochondrial damage and oxidative stress in oocytes, resulting in autophagy and apoptosis [7,13]. Currently, there are limited specific pharmacological interventions to mitigate the adverse health effect of 3-NP [14].

Recent studies suggest that dietary natural phytochemicals can alleviate 3NP toxicity. For example, berberine provides neuroprotective effects in the rat stratum against 3-NP via anti-inflammatory, antioxidant and anti-apoptosis pathways [15]. Quercetin inhibits 3-NP-induced ovarian oxidative damage by regulating autophagy through the SIR-T1/ROS/AMPK signaling pathway [16]. Silibinin, the main flavonoid compound of *Silybum marianum*, is widely recognized for its hepatoprotective properties [17,18]. In addition, silibinin also exhibits antioxidant [19], anti-aging [20], anti-inflammatory [21,22] and anti-tumor effects [23,24], making it a promising nutraceutical

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candidate. silibinin effectively scavenges ROS, inhibits lipid peroxidation, and stabilizes cell membranes, suggesting therapeutic potential for oxidative stress-related disorders [25]. In addition, it has shown promise in treating acute kidney injury [26], lung cancer [24] and cognitive dysfunction [27]. Recent studies suggest that silibinin may alleviate polycystic ovary syndrome (PCOS) by improving insulin sensitivity and regulating androgen levels, indicating a protective effect on ovarian tissue [28,29]. However, the effects of silibinin on 3-NP-induced reproductive toxicity remain largely unexplored.

Therefore, this study investigates the protective effects and mechanisms of silibinin supplementation against 3-NP-induced ovarian toxicity. Our findings revealed that silibinin alleviates 3-NP-induced oxidative stress and ovarian granulosa cell apoptosis via JNK/ERK and KEAP1/NRF2 signaling pathways. These data suggest that silibinin is a novel nutritional countermeasure with potential therapeutic value against 3-NP-induced reproductive toxicity, providing a scientific basis for developing silibinin-enriched feed additives in the animal husbandry and the veterinary industry.

2. Materials and methods

2.1. Reagents and materials

3-NP and Dimethyl Sulfoxide (DMSO) were obtained from Sigma-Aldrich (Darmstadt, Germany). Silibinin was purchased from Dalian Meilun Biotechnology Co. (Dalian, China). HiScript III RT SuperMix for qPCR (+ gDNA wiper) Kits, Trizol, and 2 × ChamQ Universal SYBR qPCR Master Mix were obtained from Nanjing Vazyme International LLC (Nanjing, China). The BCA kit was obtained from Shanghai Beyotime Biotechnology (Shanghai, China). Protein Free Rapid Blocking Buffer and the Color PAGE Gel Rapid Preparation Kit were obtained from Shanghai Epizyme Biotech Co. (Shanghai, China). Antibodies against Nrf-2, JNK, P-JNK, P38, P-P38, ERK, P-ERK were purchased from Jiangsu Affinity Biosciences Co (Changzhou, China). Antibodies against BCL2, BAX, CASPASE-3, CLEAVED-CASPASE-3, GAPDH, β -Actin, HRP-Goat Anti-Mouse IgG, HRP-Goat Anti-Rabbit IgG, the TUNEL Assay Apoptosis Detection Kit and the ECL-enhanced developer were purchased from Proteintech Group, Inc. (Rosemont, USA). Antibody against KEAP1 was obtained from Boster Biological Technology (Wuhan, China). Total Superoxide Dismutase (T-SOD) assay kit (Hydroxylamine method), Glutathione Peroxidase (GSH-PX) assay kit (Colorimetric method), Total Antioxidant Capacity (T-AOC) assay kit, Catalase (CAT) assay kit (Visible light), and Malondialdehyde (MDA) assay kit (TBA method) were obtained from NanJing JianCheng Bioengineering Institute (Nanjing, China). The DAB Kit was obtained from Beijing Zhong Jinqiao Biological Company (Beijing, China).

2.2. Animal experiment design

Ninety female ICR mice, aged 6–8-week (24 ± 2 g), were purchased from Liaoning Changsheng Biotechnology Company (Liaoning, China). Animals were raised controlled conditions ($22\text{--}24^\circ\text{C}$, 12 h light and 12 h dark cycle) with unrestricted access to water and food under. Mice were subsequently randomly divided into six groups ($n = 15$ per group): the control group (C), solvent control group (DMSO), 3-NP model group (3-NP), silibinin low-dose group (LSB), silibinin middle-dose group (MSB), and silibinin high-dose group (HSB). The experiment lasted 14 days. 3-NP was dissolved in saline, while silibinin was solubilized with dimethyl sulfoxide (DMSO) following the instructions from the manufacturer. According to the previous methods [9,30], the experimental groups were treated as follows: the control group was intraperitoneally injected of saline (200 μL , I.P) twice daily from 7 to day 14. The 3NP group was intraperitoneally injected with 12.5 mg/kg 3-NP (200 μL , I.P) twice daily from day 7 to day 14. The silibinin group was pretreated with silibinin (100/150/200 mg/kg) by oral gavage (100 μL , gavage) daily for 14 days, with concurrent intraperitoneally injection of 12.5 mg/kg 3-NP

(200 μL , I.P) twice daily from day 7 to day 14. The solvent control group was administered 100 μL of DMSO (100 μL , gavage) by oral gavage daily for 14 days, with concurrent intraperitoneally injection of saline (200 μL , I.P) twice daily from day 7 to day 14. All chemical solutions were prepared fresh daily and administered at consistent time intervals. The Shenyang Agricultural University Animal Care and Use Committee approved all animal experimental procedures (No. 24121201).

2.3. Sample collection

Following completion of all treatments, animals were fasted for 12 h and weighed. Mice were then anesthetized with 1 % sodium pentobarbital and euthanized by cervical dislocation. Blood, livers, ovaries, kidneys, and hearts were quickly collected. Serum was obtained by centrifugations and stored at -80°C . Livers, hearts, ovaries, and kidneys were weighed, and organ mass index (%) was calculated as follows: livers (ovaries, kidneys, and hearts) mass/body mass $\times 100$. Both ovaries were collected: one was fixed with 4 % paraformaldehyde, and the other one was immediately preserved at -80°C for subsequent assays.

2.4. Detection of oxidative stress index in the serum

The serum samples were collected to evaluate oxidative stress-related indices in the serum. The serum levels of T-SOD, GSH-Px, CAT, T-AOC, and MDA in each group were determined using commercially available kits according to the manufacturers' instructions. All measurements were performed in six biological replicates in each group.

2.5. Histological evaluation

2.5.1. Hematoxylineosin staining and follicle counting

Ovaries were fixed overnight at 4°C in 4 % paraformaldehyde, dehydrated, and embedded in paraffin. The paraffin-embedded ovaries were then sectioned at 4–5 μm , and stained with hematoxylin and eosin (H&E). All slides were examined under the microscope (Leica, CMS GmbH D-35578, Wetzlar) for follicle counting as previously described [9]. Briefly, every fifth section of the ovary was examined under the microscope. To avoid multiply counting the same follicle, only follicles with a visible oocyte nucleus were included in the count. The number of primordial follicles was assessed in all the serial sections of an ovary. The follicles were categorized as follows according to previous method [9]: primordial follicle: an oocyte surrounded by a layer of flattened granulosa cells; primary follicle: an oocyte surrounded by one or two intact layers of cuboidal granulosa cells; secondary follicle: an oocyte surrounded by multiple layers of cuboidal granulosa cells but lacking a visible follicular lumen; mature follicle: an oocyte surrounded by several layers of cuboidal granulosa cells, with a clear follicular lumen present; atretic follicle: an oocyte with a degenerated nucleus, characterized by chromosomal and cytoplasmic breakdown, with a reduction in the granulosa layer and hypertrophy of the follicular membrane cells. The corpus luteum is found in the ovarian cortex, covered by a connective tissue peritoneum and distinctly separated from the surrounding tissue. The follicle rate indicates the proportion of follicles at different developmental stages in relation to the total number of follicles. All experiments were independently performed in three biological replicates in each group.

2.5.2. Immunohistochemistry (IHC) assays

The IHC assays of ovary sections were performed following previously established protocols [31]. Briefly, Paraffin-embedded ovaries were sectioned at 4–5 μm thickness. Sections were incubated with anti-Caspase-3 (1:300) primary antibody overnight at 4°C . Subsequently the slides were incubated with horseradish peroxidase-labeled rabbit anti-goat IgG secondary antibody (1:1000) at room temperature for 3 h. Samples were then stained and labeled with 3,3'-diaminobenzidine (DAB), and the slides were visualized and captured with a microscope

(Leica, CMS GmbH D-35578, Wetzlar). All experiments were independently performed in three biological replicates in each group.

2.6. Total RNA extraction and qPCR

Total RNA from ovarian tissue was obtained using the Trizol Total RNA method. RNA quality and concentration were analyzed at 260 nm and 280 nm wavelength via spectrophotometry (ASUV-1100, AS ONE, China). RNA of sufficient quality (A260/A280: 1.8–2.1) was utilized for subsequent RT-qPCR analysis. The purified total RNA (1 µg) as a template for cDNA synthesis was then reverse transcribed using reverse transcriptase in a T100™ Thermal Cycler System (621BR20114, BIO-RAD, Singapore) for subsequent experiments. Real-time quantitative polymerase chain reaction (RT-qPCR) using the qTOWER³G system (D-07745, Jena, Germany) with 2 × ChamQ Universal SYBR qPCR Master Mix (Vazyme, China) was performed. RT-qPCR conditions were as follows: 95 °C for 2 min, followed by 40 cycles of 95 °C for 15 s and 60 °C for 1 min. Relative mRNA quantification was performed using the 2^{-ΔΔCT} method. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was used as a reference gene to normalize relative RNA expression levels. The sequences of specific primers designed using Primer 5.0 are detailed in Table 1. All experiments were independently performed in three biological replicates in each group.

2.7. Western blot analysis

Total protein was extracted from snap frozen ovarian tissues stored at – 80 °C using RIPA cell lysate. Protein concentration were quantified with BCA Kits. Protein samples were separated by electrophoresis in polyacrylamide gels using POWERPAC™-HC Basic Electrophoresis (043BR59326, BIO-RAD, Singapore) and transferred to PVDF membranes. Membranes were blocked for 1 h at room temperature with rapid protein-free blocking buffer. Subsequently, membranes were incubated at 4 °C overnight with target primary antibodies: BAX(1:1500), BCL2 (1:3000), CASPASE-3(1:2000), NRF2(1:1000), ERK(1:1500), P-ERK (1:1000), JNK(1:1000), P-JNK(1:1000), KEAP1 (1:1500), GAPDH (1:5000), β-ACTIN(1:10000), P38(1:1000), P-P38(1:1000). After washing with TBST, membranes were then incubated with horseradish peroxidase (HRP)-coupled secondary antibodies (1:5000) for 2 h at room temperature. Finally, protein bands were visualized using a UVP ChemStudio™ image analyzer (J113258, Jena, Upland, USA), and quantified using ImageJ software (version 2.14.0, Bethesda, USA). All experiments were independently performed in three biological replicates in each group.

2.8. Statistical analyses

All experiments were independently performed at least three times. All statistical analyses were performed using SPSS (version 26.0,

Chicago, IL, USA). All data were assessed for significance using one-way (ANOVA) analysis of variance followed by LSD post hoc tests for multiple comparisons. When the data did not satisfy normal distribution, a nonparametric, Kruskal-Wallis (H) test was used as an alternative method. Graphs were generated by GraphPad Prism (version 8.0, California, USA). Results are expressed as mean ± standard error of the mean (SEM). Significant results were defined as $p^* \leq 0.05$ and $p^{**} \leq 0.01$.

3. Results

3.1. Silibinin significantly ameliorates 3-NP-induced ovarian index decline

The chemical structure of silibinin is illustrated in Fig. 1A. In order to evaluate the protective effects of silibinin against ovarian damage induced by environmental toxicants, we established a 3-NP exposure model (Fig. 1B). While 12.5 mg/kg 3-NP did not alter either body weight or the indices of internal organs (liver, heart, and kidney, Fig. 1C–F), it specifically induced the ovarian index compared to the control group (Fig. 1G). Notably, silibinin treatment dose-dependently recovered the ovarian index (Fig. 1G), indicating a protective effect, specifically against 3-NP-induced ovarian damage.

3.2. Silibinin reduces 3-NP-induced follicular atresia

We performed a histopathological examination using H&E staining to assess ovarian structure changes across different groups (Fig. 2A). In the control group ovaries displayed intact follicular structures with few atretic follicles, whereas 3-NP exposure significantly increased follicular atresia. Remarkably, silibinin treatment dose-dependently reversed this pathological condition in comparison with the 3-NP group (Fig. 2B–G). These findings indicate that silibinin eliminates 3-NP-induced reproductive damage.

3.3. Silibinin reduces 3-NP-induced oxidative stress

Since 3-NP-associated ovarian damage is often accompanied by oxidative stress, and we measured serum activities of T-SOD, GSH-Px, CAT, and T-AOC, along with the content of MDA, to investigate the potential regulation of oxidative stress by silibinin. The results elucidated that exposure to 3-NP significantly suppressed the activities of these antioxidant enzyme in serum, and markedly increased MDA content compared to the control group (Fig. 3A–E). Conversely, silibinin treatment dramatically increased T-SOD, GSH-Px, CAT and T-AOC activities in serum, and decreased MDA content compared with the 3-NP group (Fig. 3A–E).

In addition, qPCR analysis of ovarian tissues revealed that compared to the control group, 3-NP exposure remarkably upregulated *Gpx1*, *Sod2* and *Cat* mRNA expressions by 210 %, 260 %, and 190 %, respectively (Fig. 3F–H). However, silibinin treatment significantly reduced the expression of these mRNA compared to the 3-NP group, particularly in the HSB group which downregulated *Gpx1*, *Sod2* and *Cat* mRNA by 71 %, 68 % and 66 %, respectively (Fig. 3F–H), suggesting that silibinin enhanced endogenous antioxidant capacity. Our data indicated that silibinin has the potential to function as a potential therapeutic redox modulator.

3.4. Silibinin attenuates 3-NP-induced ovarian granulosa cell apoptosis

To assess the protective effects of silibinin against 3-NP-induced ovarian apoptosis, we analyzed the expressions of apoptosis-related genes using qPCR and Western blot. The qPCR results demonstrated that compared to the control group, 3-NP exposure dramatically upregulated the mRNA of the pro-apoptotic gene *Bax* and *Caspase 3* by 160 % and 70 %, downregulated the anti-apoptotic gene *Bcl-2* mRNA by 44 %, and

Table 1
The primer sequences.

Genes	Primer sequences (5'-3')
<i>GADPH</i>	Forward: GGT TGT CTC CTG CGA CTT CA Reverse: GGG TGG TCC AGG GTT TCT TA
<i>Caspase-3</i>	Forward: AGA GGA ATG ATT GGG GGT G Reverse: TTG CTA GGC AGT GGT AGC G
<i>Bax</i>	Forward: TTT CAT CCA GGA TCG AGC AGG Reverse: GCA AAG TAG AAG AGG GCA ACC A
<i>Bcl-2</i>	Forward: CTAC CGT CGT GAC TTC GCA Reverse: TAA CCC AGC CTC CGT TAT CC
<i>Sod2</i>	Forward: GTG GGA GTC CAA GGT TCA GG Reverse: CAT TCT CCC AGT TGA TTA CAT TCC
<i>Gpx1</i>	Forward: TGC GAG TGA ATG GTG AGA AGG Reverse: TGG GAC AGC AGG GTT TCT ATG T
<i>Cat</i>	Forward: TCT TGT TCA GTG ACC GAG GGA Reverse: GCC CTG GTC GGT CTT GTAATG

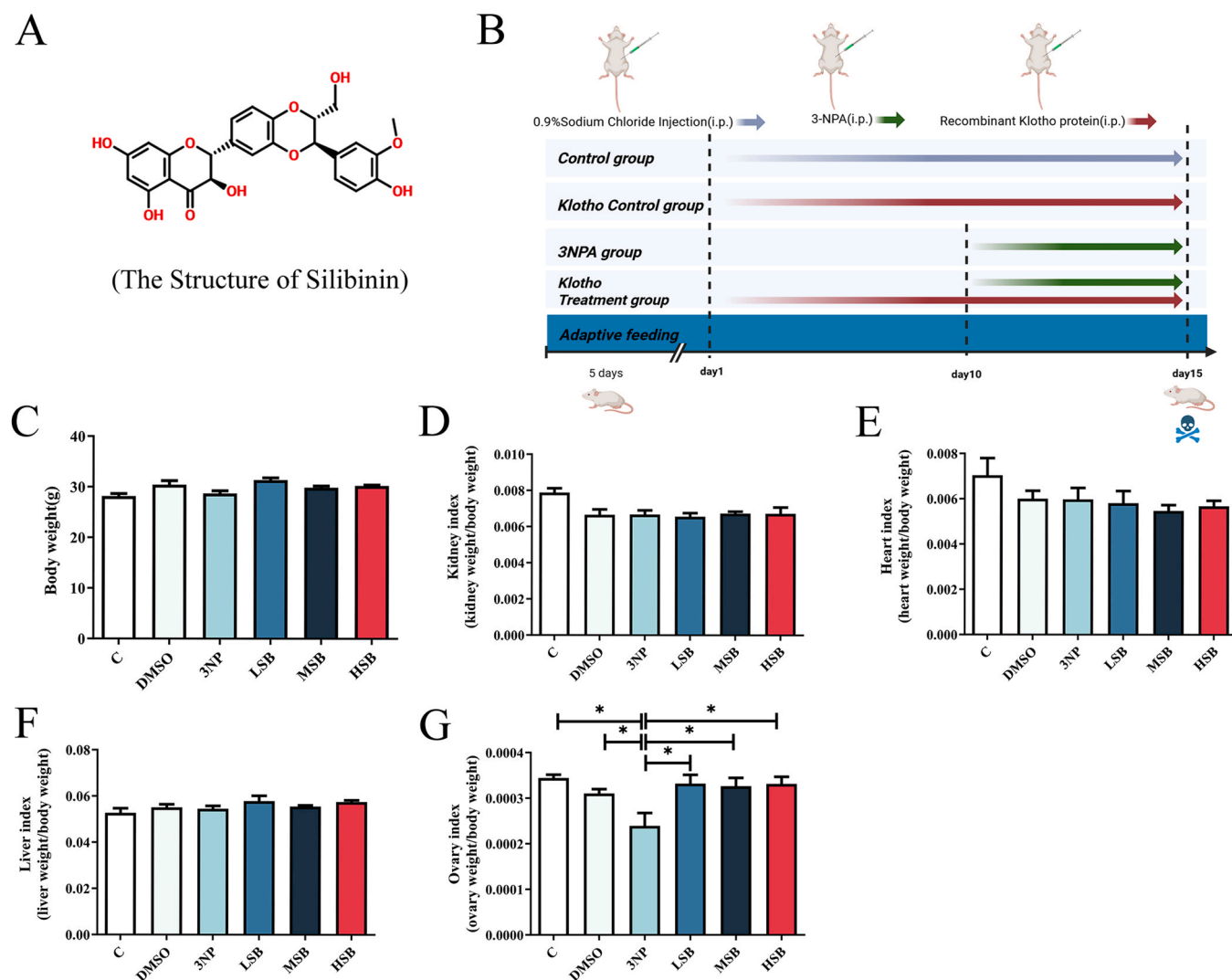


Fig. 1. Silibinin significantly increases the ovarian index in mice induced with 3-NP. (A) Schematic structure of silibinin; (B) Schematic diagram of experimental model establishment; (C) Body weight of mice; (D) Renal organ indexes; (E) Cardiac organ indexes; (F) Hepatic organ indexes; (G) Ovarian organ indexes. The *p*-values were calculated using Kruskal-Wallis or ANOVA (post-LSD) (*n* = 5). The statistics of the data are presented in the form of histograms and expressed as mean ± standard error of the mean (SEM). Significant results were reported as $p^* \leq 0.05$ and $p^{**} \leq 0.01$.

and reduced the Bcl-2/Bax expression ratio by 78 % (Fig. 4A–D). At the same time, compared to 3-NP group, the silibinin treatment dose-dependently reversed these alterations, decreasing *Bax* and *Caspase3* mRNA expressions and increasing *Bcl2* mRNA and *Bcl-2/Bax* expression ratios. In the HSB group, silibinin exhibited the most significant anti-apoptotic effect, downregulating *Bax* and *Caspase3* mRNA by 53 % and 44 %, while upregulating *Bcl-2* mRNA and the *Bcl-2/Bax* ratio by 120 % and 370 %, respectively. (Fig. 4A–D).

Western blot analysis confirmed these findings at the protein levels. Compared to the control group, 3-NP exposure significantly increased ovarian expression of BAX, CASPASE-3, and CLE-CASPASE-3 protein by approximately 70 %, 100 %, and 50 %, respectively, while reducing BCL2 protein and the BCL2/BAX ratio by 40 % and 64 %, respectively. Conversely, silibinin treatment dramatically attenuated these effects compared to the 3-NP group, decreasing the expressions of BAX, CASPASE-3, and CLE-CASPASE-3 protein, and increasing BCL2 protein levels and the BCL2/BAX expression ratio. The most significant alleviating effects were observed in the HSB group, where silibinin down-regulated BAX, CASPASE-3, and CLE-CASPASE-3 proteins by approximately 54 %, 47 %, and 48 %, respectively, and increased BCL2 protein levels and the BCL2/BAX expression ratio by 70 % and 270 % (Fig. 4E–K). Immunohistochemistry revealed that CASPASE3 was

predominantly exhibited in the granulosa cells (Fig. 4L). Collectively, our results indicated that silibinin protects against 3NP-induced apoptosis in ovarian granulosa cells.

3.5. Silibinin alleviates 3-NP-induced apoptosis in granulosa cells through the KEAP1/NRF2 pathway

To determine whether the KEAP1/NRF2 signaling pathway is involved in the protection against 3-NP-induced oxidative stress by silibinin, we assessed KEAP1 and NRF2 proteins expression levels via Western blot compared to the control group, exposure to 3-NP significantly increased KEAP1 protein levels by 80 %, while simultaneously reducing NRF2 protein expression by 41 % (Fig. 5A–D). Notably, compared to the 3-NP group, silibinin treatment dose-dependently reversed these alterations, decreasing KEAP1 protein expression and increasing NRF2 protein expression. The most significant effects were observed in the HSB group, where silibinin reduced KEAP1 protein levels by 62 % and elevated NRF2 protein by 40 % (Fig. 5A–D). These results demonstrated that silibinin activates cellular antioxidant defenses by modulating the KEAP1/NRF2 pathway.

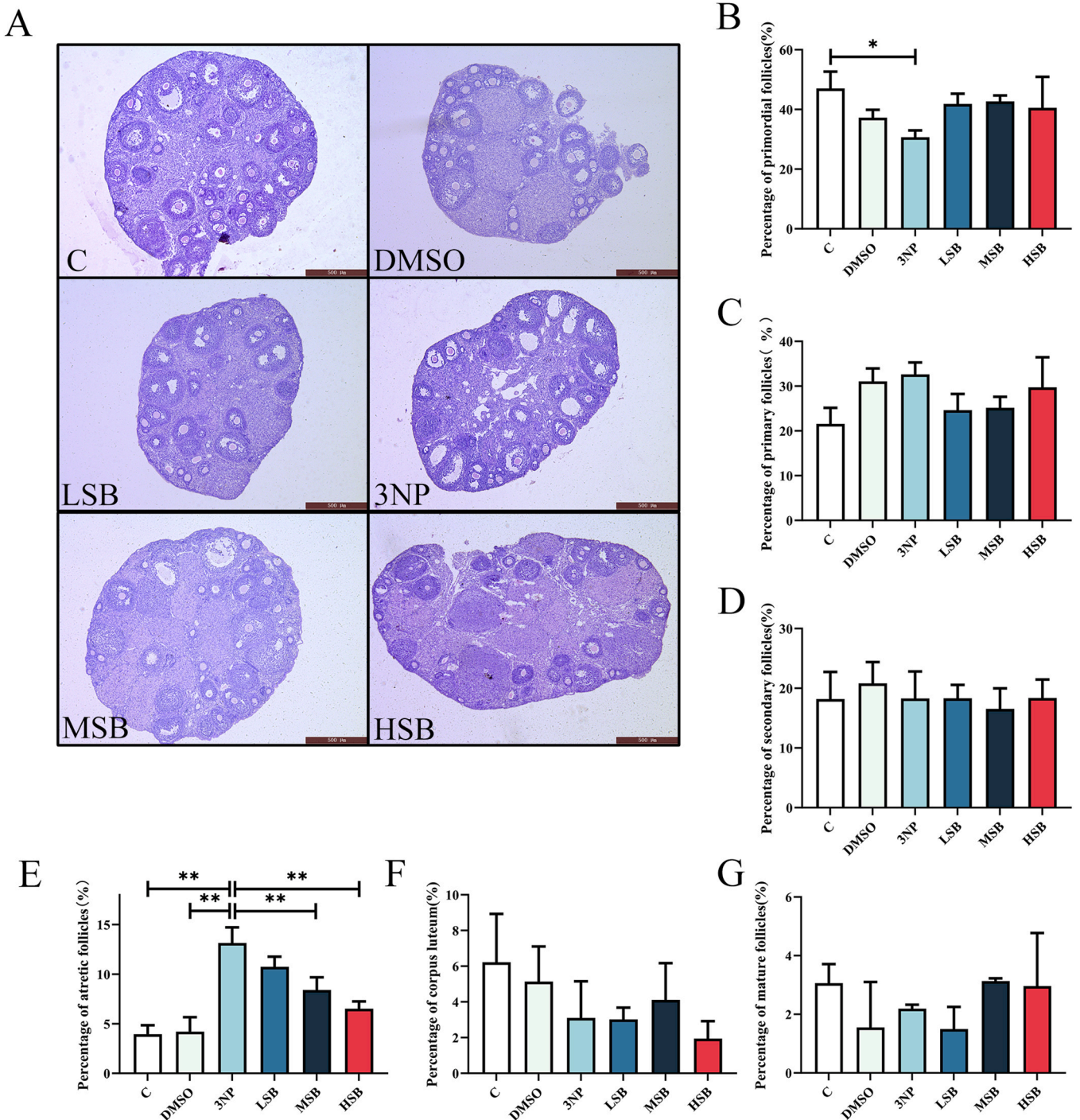


Fig. 2. Silibinin reduces 3-NP-induced follicular atresia: (A) Representative images of HE-stained Ovary tissues showing histopathological changes. (B) Proportion statistic of primordial follicles in each group; (C) Proportion statistic of primary follicles in each group; (D) Proportion statistic of secondary follicles in each group; (E) Proportion statistics of atretic follicles in each group; (F) Proportion statistics of corpus luteum in each group; (G) Proportion statistics of mature follicles in each group. Scale bars have been identified in the pictures. The p-values were calculated using Kruskal-Wallis or ANOVA (post-LSD) ($n = 3$). These results are expressed as mean $\pm$ standard error of the mean (SEM). Significant results were reported as $p^* \leq 0.05$ and $p^{**} \leq 0.01$.

3.6. Silibinin modulates 3-NP-induced apoptosis in granulosa cells via the MAPK signaling pathway

To further investigate the molecular mechanisms of silibinin in alleviating 3-NP-induced apoptosis in the granulosa cells, we examined the activation of the MAPK pathway by Western blot. Our results demonstrated that compared with the control group, 3-NP exposure dramatically increased the phosphorylation of both JNK and ERK by

180 % and 60 %, respectively (Fig. 6 A–C). In contrast to the 3-NP group, silibinin treatment markedly attenuated JNK and ERK phosphorylation in the ovarian tissues. The effect was most obvious in the HSB group, where silibinin reduced phosphorylation of JNK and ERK by 47 % and 41 %, respectively (Fig. 6A–C). Notably, there was no significant change in the P38 signaling pathway (Fig. 6A, D). These results indicated that silibinin protected against 3-NP-induced granulosa cell apoptosis by modulating the JNK/ERK signaling pathway.

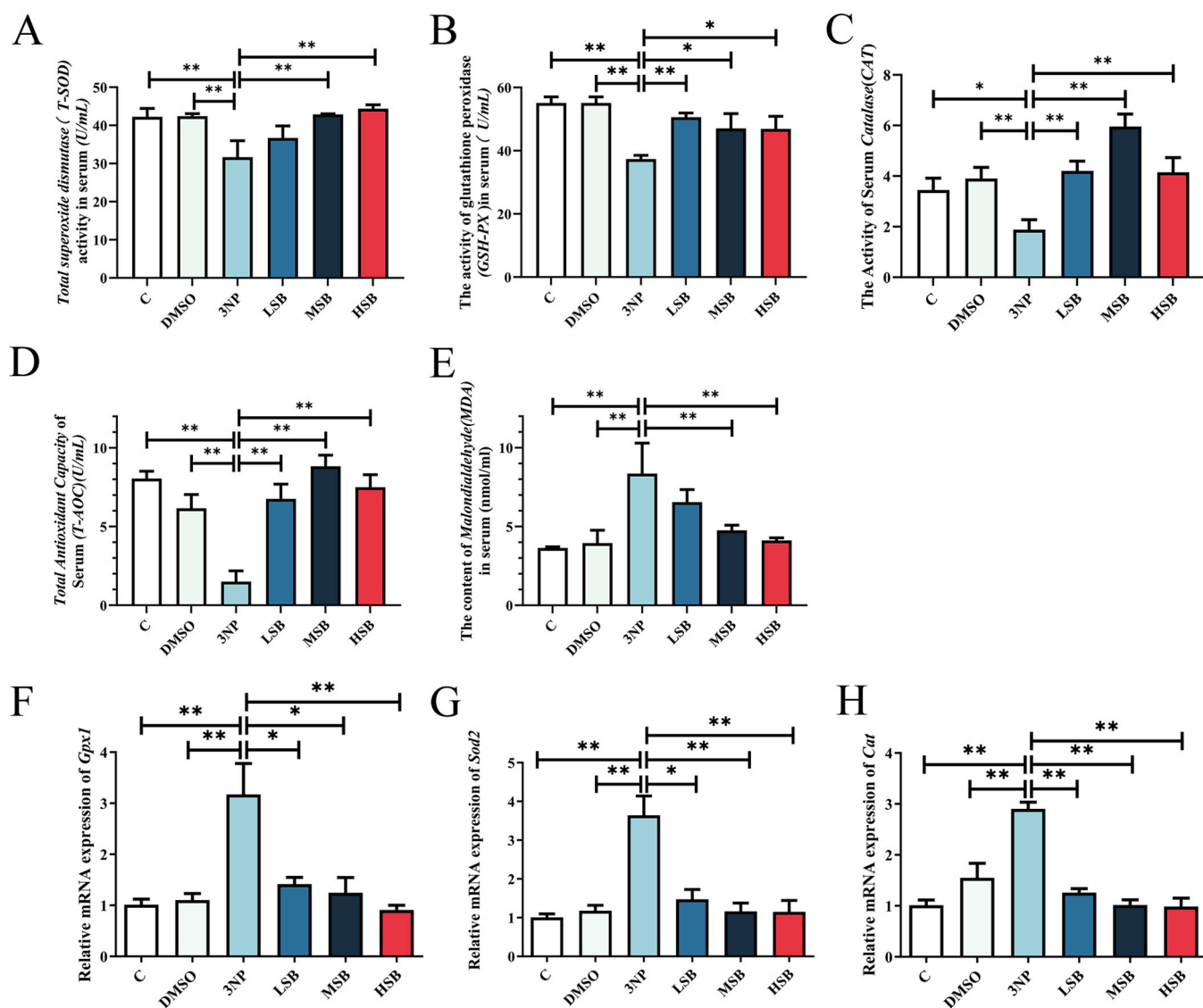


Fig. 3. Silibinin reduces 3NP-induced oxidative stress: (A) Total Superoxide Dismutase activity in serum of mice; (B) Glutathione Peroxidase activity in serum of mice; (C) Catalase activity; (D) Total Antioxidant Capacity in serum of mice; (E) The content of Malondialdehyde in serum of mice (F) mRNA levels of *Gpx1* in mouse ovaries; (G) mRNA levels of *Sod2* in mouse ovaries (H) mRNA levels of *Cat* in mouse ovaries. The p-values were calculated using Kruskal-Wallis or ANOVA (post-LSD) (A–D, n = 6; F–H, n = 3). Significant results were reported as $p^* \leq 0.05$ and $p^{**} \leq 0.01$.

4. Discussion

Growing evidence indicates that mitochondrial dysfunction caused by the mycotoxin 3-NP induces oxidative stress, which can lead to reproductive toxicity, including granulosa cells apoptosis and follicular atresia [9,32]. Silibinin, a naturally occurring flavonoid, is recognized for its diverse biological properties, including its antioxidant capacity and protective effects on ovarian health [28,29]. Notably, previous investigations have demonstrated that silibinin has neuroprotective potential against 3-NP-induced oxidative damage to neural tissue [33]. In the present research, we utilized 3-NP to establish an animal model of oxidative stress, aiming to investigate whether silibinin exerts potential protective effects on ovaries exposed to 3-NP.

3-NP, a mycotoxin derived from the fungus *Arthrinium*, naturally occurs in spoiled or moldy sugarcane and certain legumes, such as Ruo leaf and Astragalus. The consumption of agricultural products contaminated with 3-NP poses serious health risks, potentially inducing systemic toxicity and neuromotor impairments in mammals, raising significant food safety concerns. Recent toxicological research has identified 3-NP as a reproductive risk factor, associated with granulosa

cells apoptosis [32], the arrest of oocyte maturation, and follicular atresia [9]. In the current research, we used an animal model to evaluate 3-NP-induced ovarian injury. Our findings reveal that supplementation of 12.5 mg/kg of 3-NP in mice specifically resulted in a significant reduction in ovarian organ index and accelerated follicular atresia, confirming ovarian toxicity consistent with previous reports [9]. Furthermore, our subsequent research demonstrated that exogenous silibinin application reduced follicular atresia in a dose-dependent manner and exerted a protective role against 3-NP-induced ovarian damage. It is well-known that silibinin has attracted considerable scientific attention due to its multiple bioactive properties such as hepatoprotective, antioxidant, and anti-inflammatory effects. Recently it has also been reported that silibinin alleviates PCOS by enhancing insulin sensitivity, modulating androgen biosynthesis, and reducing oxidative stress [28,29]. Taken together, our findings suggest that silibinin, a plant-extracted flavonoid, is a potential phototherapeutic candidate for ameliorating female reproductive disorders.

An antioxidant network exists within the ovarian microenvironment, primarily located in the follicular fluid, granulosa cells and oocytes. This integrated system, comprising endogenous enzymatic antioxidants such

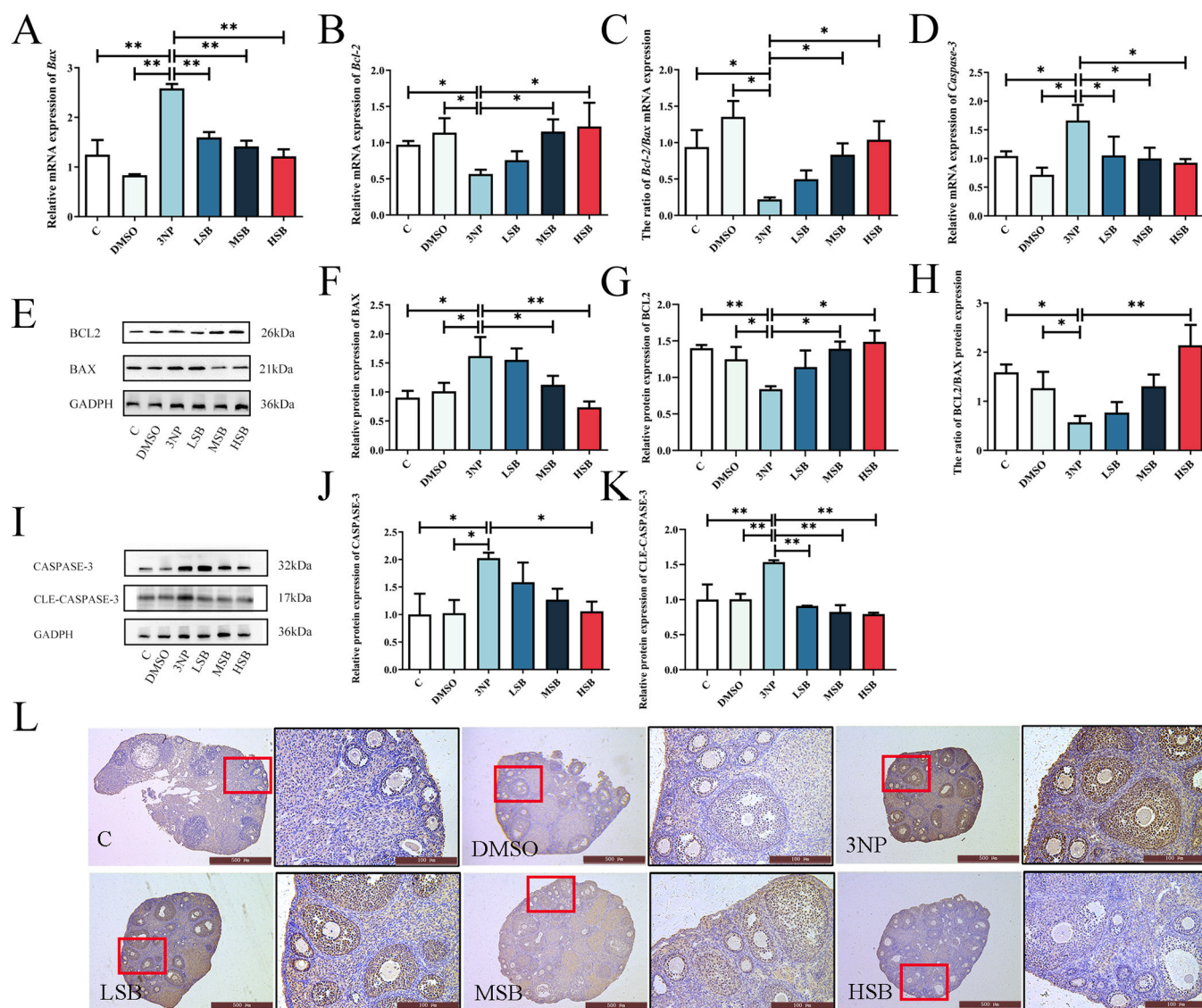


Fig. 4. Silibinin inhibits 3-NP-induced ovarian cell apoptosis: (A) mRNA levels of *Bax* in mouse ovaries; (B) mRNA levels of *Bcl-2* in mouse ovaries; (C) The ratio of *Bcl-2/Bax* mRNA expression; (D) mRNA levels of *Caspase-3* in mouse ovaries; (E) The protein expression of Bcl-2 and Bax in mouse ovaries; (F): Quantification of BAX protein levels; (G) Quantification of BCL2 protein levels; (H) Quantitative ratio of BCL2/BAX protein levels; (I): The protein expression of CASPASE-3 and CLEAVED-CASPASE-3 in mouse ovaries; (J): Quantification of CASPASE-3 protein levels; (K): Quantification of CLEAVED-CASPASE-3 protein levels; (L): Localization of CASPASE-3 in mouse ovaries by immunohistochemistry (4 ×, 40 ×). Histograms are expressed as gray-scale values of quantified protein bands. The p-values were calculated using Kruskal-Wallis or ANOVA (post-LSD) (n = 3). These results are expressed as mean ± standard error of the mean (SEM). Significant results were reported as $p \leq 0.05$ and $p^{**} \leq 0.01$.

as T-SOD, GSH-Px, CAT, and T-AOC, as well as non-enzymatic antioxidant substances, plays essential roles in counteracting the overproduction of ROS and maintaining follicular homeostasis. Accumulating evidence indicates that oxidative stress-induced lipid peroxidation and DNA damage are crucial mediators of ovarian dysfunction [9,34]. As a supplement to endogenous defenses, dietary antioxidants including melatonin, resveratrol, vitamin C, and vitamin E have been reported to scavenge excess ROS, offering therapeutic potential for ovarian oxidative damage [35–38]. Our experimental results reveal that 3-NP dramatically decreased the activity of T-SOD, GSH-Px, CAT, and T-AOC in serum, while increasing MDA content, consistent with previous studies on 3-NP-induced oxidative damage [39,40]. Conversely, silibinin supplementation effectively reversed these alterations by enhancing antioxidant enzyme activities and reducing MDA production, suggesting its significant antioxidative properties. These findings align with previous research demonstrating that silibinin alleviated 3-NP-induced neurotoxicity in male mice by reducing oxidative

stress and neuroinflammation [33]. To further investigate, we assayed *Gpx1*, *Sod2* and *Cat* mRNA expressions in ovarian tissues. Our results demonstrated that 3-NP upregulated the expressions of *Gpx1*, *Sod2* and *Cat* mRNA, indicating a compensatory antioxidant response to oxidative challenge. This observation aligns with previous reports documenting adaptive mechanisms in the ovaries in response to 3-NP [9]. Interestingly, supplementation with silibinin normalized these transcriptional alterations, implying its ability to alleviate oxidative stress through neutralizing ROS rather than compensatory gene activation. These data suggest that silibinin is a promising antioxidant to protect ovaries against oxidative damage.

ROS plays crucial roles in maintaining follicular homeostasis by mediating the dynamic balance between follicular proliferation and atresia [41]. However, overproduction of ROS disrupts the ovarian redox balance, compromising the antioxidant defense and initiating oxidative damage cascade. This pathophysiological state promotes the apoptosis of granulosa cells and oocytes, which is the critical factor in

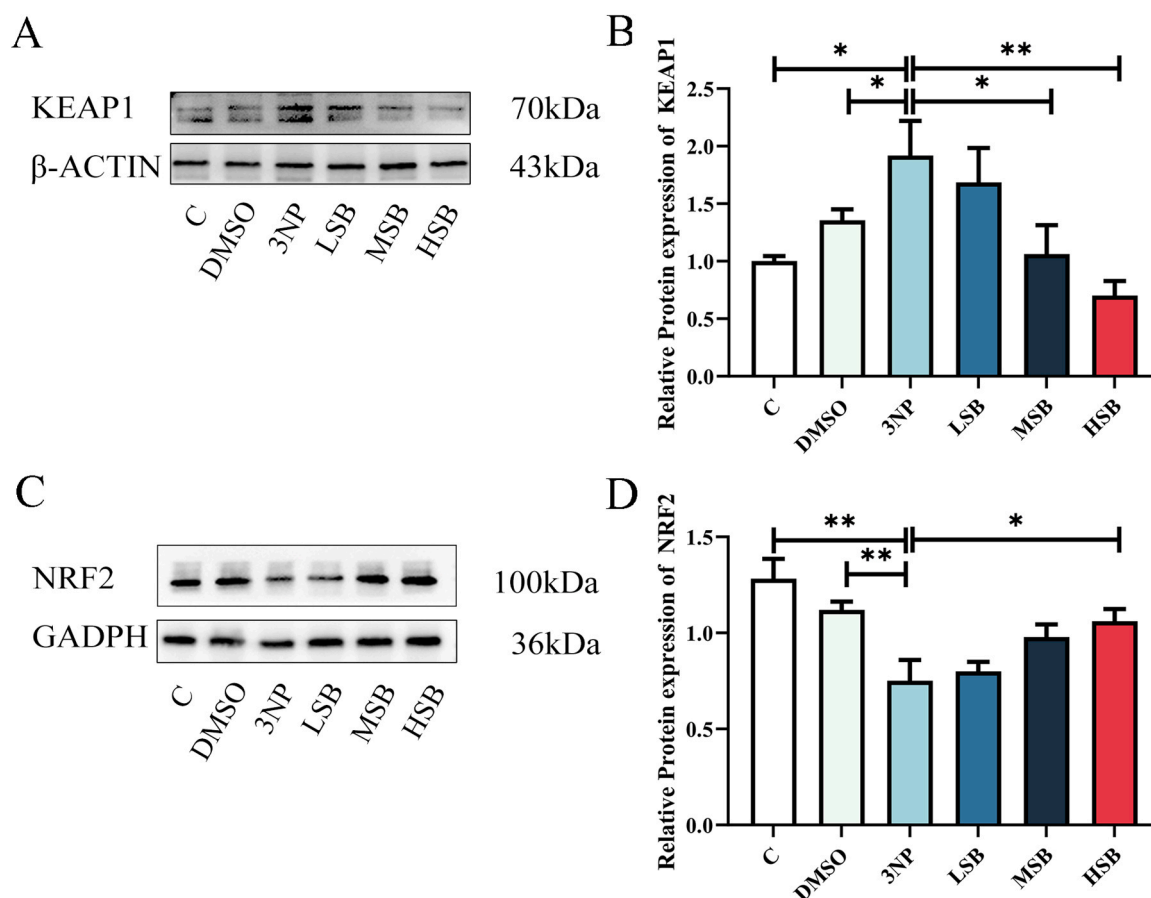


Fig. 5. Silibinin attenuates 3-NP-induced granulosa cell apoptosis via the KEAP1/NRF2 pathway: (A) The protein expression of KEAP1 in mouse ovaries; (B): Quantification of KEAP1 protein levels (C) The protein expression of NRF2 in mouse ovaries; (D): Quantification of NRF2 protein levels. Histograms are expressed as gray-scale values of quantified protein bands. The p-values were calculated using Kruskal-Wallis or ANOVA (post-LSD) ($n = 3$). These results are expressed as mean $\pm$ standard error of the mean (SEM). Significant results were reported as $p^* \leq 0.05$ and $p^{**} \leq 0.01$.

follicular atresia [42]. Notably, it has been widely reported that food-borne contaminants including mycotoxins (for instance, aflatoxin B1), heavy metals (for instance, Cd, Pb), and other persistent organic pollutants in the environment have been epidemiologically associated with ovarian oxidative stress and follicular atresia [43–46]. It is well known that follicular atresia is initiated by the excessive apoptosis of granulosa cells. Meanwhile, increasing evidence suggests that silibinin regulates the apoptosis of cancer cells through multiple signaling pathways, including those involved in glioblastoma, gastric and colon cancer cells [47–50]. Recently, it has been reported that silibinin attenuated 5-fluorouracil (5-FU)-induced ovarian apoptosis [51], which prompted our investigation into its potential against 3-NP-induced ovarian toxicity. Our results confirm that 3-NP significantly increased apoptosis in ovarian granulosa cells, consistent with a model of 3-NP-induced ovarian injury [7,39,52]. Crucially, silibinin supplementation alleviated this effect by downregulating pro-apoptotic gene expression (*Bax*, *Caspase-3*) while simultaneously increasing the transcription of the anti-apoptotic gene *Bcl-2*. Our data indicate that the multi-functional phytochemical silibinin provides ovarian protection by reducing granulosa cell apoptosis and decreasing follicular atresia.

The KEAP1/NRF2 signaling pathway is an essential modulator of cellular redox homeostasis, coordinating protective responses against oxidative stress by activating cytoprotective genes transcription [53, 54]. Under physiological conditions, Keap1 binds Nrf2, promoting its cytoplasmic degradation, maintaining basal antioxidants levels. Upon exposure to ROS and other harmful stressors, Keap1-mediated repression is alleviated via structural modifications, allowing Nrf2 nucleus translocation. Nuclear Nrf2 binds to antioxidant response element

(ARE), triggering downstream antioxidants genes activation to scavenge excess ROS and enhance antioxidant capacity [53,55,56]. Increasing evidences indicates that dietary phytochemicals exhibit therapeutic potential in modulating the KEAP1/NRF2 pathway. For example, research has shown that resveratrol protects rat granulosa cells against H_2O_2 through the SIRT1/Nrf2/ARE signaling pathway [57]. Carnosol alleviates PCOS via Keap1/Nrf2/HO-1 signaling pathway [58]. Genistein functions as a therapeutic factor in PCOS through the ER-Nrf2-Foxo1-ROS pathway [59]. Silibinin also modulates the KEAP1/NRF2 pathway to reduce oxidative stress [60–62]. In the current study, 3-NP notably elevated KEAP1 protein expression while suppressing NRF2 protein expression in the ovary. In particular, silibinin intervention effectively reversed these protein expression changes, consistent with previous research [60–62]. Our results demonstrate that silibinin significantly ameliorated 3NP-induced ovarian dysfunction via KEAP1/NRF2 pathway activation. While our data position silibinin as a promising candidate for therapeutic intervention against ovarian dysfunction induced by environmental toxins, the precise molecular target of silibinin in the KEAP1/NRF2 pathway requires further investigation.

Members of the mitogen-activated protein kinase (MAPK) family are widely recognized as important regulators of cellular homeostasis, and play essential roles in various biological processes including oxidative stress, cell proliferation, differentiation and apoptosis [63–65]. Specifically, JNK signaling predominantly drives apoptotic pathways by inhibiting anti-apoptotic protein Bcl-2/Bcl-xL and enhancing the activity of pro-apoptotic Bax [66]. Meanwhile, the ERK signaling pathway plays critical roles in survival, differentiation, and antioxidant defense

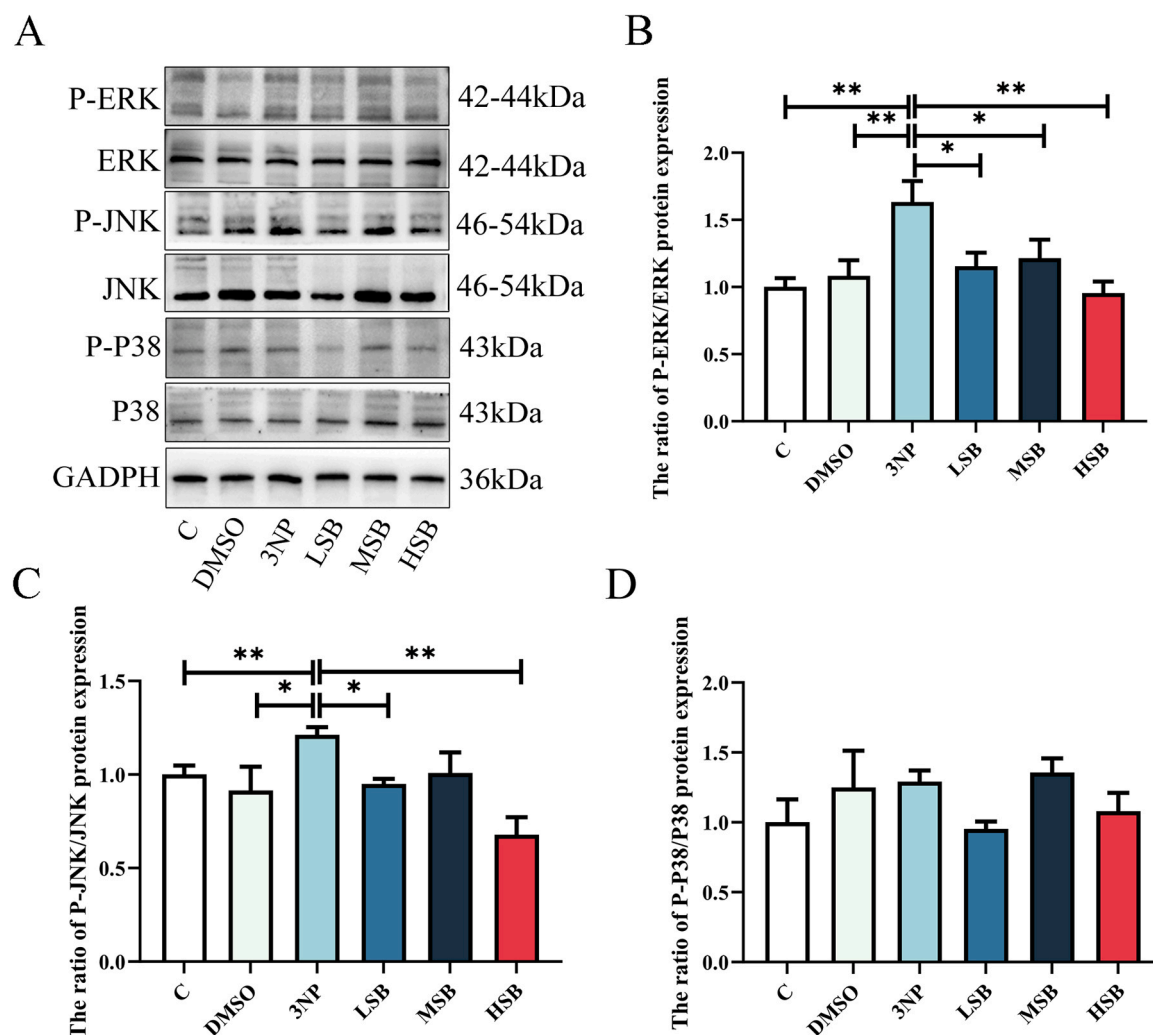


Fig. 6. Silibinin regulates 3-NP-induced granulosa cell apoptosis through the MAPK signaling pathway in mice: (A) The protein expression of P-JNK, JNK, P-ERK, ERK, P-P38 and P38 in mouse ovaries; (B) Quantification of P-ERK/ERK protein levels; (C): Quantification of P-JNK/JNK protein levels; (D): Quantification of P-P38/P38 protein levels. Histograms are expressed as gray-scale values of quantified protein bands. The p-values were calculated using Kruskal-Wallis or ANOVA (post-LSD) ($n = 3$). These results are expressed as mean $\pm$ standard error of the mean (SEM). Significant results were reported as $*p \leq 0.05$ and $**p \leq 0.01$.

mediated through Nrf2 nuclear translocation [64]. Notably, phosphorylation events in the JNK/ERK cascade have acted as key mediators of various apoptotic processes related to oxidative stress [63,67,68]. Intriguingly, dietary antioxidants such as VE and lycopene regulate fluorosis-induced ovarian damage by inhibiting JNK/ERK phosphorylation [69]. Similarly, tetrahydrocurcumin, a bioactive curcumin metabolite, effectively alleviate oxidative stress and cardiomyocyte apoptosis against lipopolysaccharide by modulating the JNK/ERK signaling pathways [70]. Our research demonstrated that exposure to 3-NP dramatically suppressed the phosphorylation levels of both JNK and ERK, which is a pathological signature correlated with oxidative damage and the activation of apoptosis. Silibinin supplementation normalized the kinase activation pattern, which is consistent with the therapeutic effects of antioxidants observed [69,70]. Our data indicated that silibinin is a potential dietary intervention for alleviating ovarian dysfunction induced by environmental toxins by targeting JNK/ERK signaling pathway.

5. Conclusions

In conclusion, the present study revealed that silibinin significantly alleviated 3-NP-induced oxidative stress and granulosa cell apoptosis in the ovary by inhibiting the JNK/ERK signaling pathway and activating

the KEAP1/NRF-2 signaling pathway. Our findings indicate that silibinin might be a potential multi-target candidate therapeutic agent against 3-NP-induced reproductive toxicity, providing a scientific theoretical basis for the development of silibinin-enriched feed additives for the livestock and veterinary industries.

CRediT authorship contribution statement

gai yedan: Writing – original draft, Visualization, Validation, Resources, Project administration, Funding acquisition, Conceptualization. **Wenhao Wu:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Haoyu Wang:** Methodology, Investigation, Data curation. **Jianmin Hu:** Writing – review & editing, Supervision, Project administration. **Xinhong Luan:** Writing – review & editing, Supervision, Project administration. **Yuqing Li:** Software. **Changbo Li:** Software. **Yingyu Wang:** Funding acquisition, Formal analysis. **Yaolu Zhao:** Writing – review & editing, Validation.

Ethics Statement

The animal study was reviewed and approved by the Ethics Committee of the Shenyang Agricultural University under permit No. 24121201.

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Conflicts of Interest

The authors declare no conflict of interest.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yedan Gai reports financial support was provided by the National Natural Science Foundation of China. Yingyu Wang reports financial support was provided by China Postdoctoral Science Foundation. Yedan Gai reports financial support was provided by the Liaoning Provincial Department of Education funding for fundamental research projects. Yedan Gai reports financial support was provided by the Liaoning Provincial Department of Science and Technology funding for Joint research projects. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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